

ELTAHER EL-SHANTA

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PROFESSIONAL SUMMARY

- Over 30 years accumulated experience performing systems analysis, design, implementation, testing, deployment, and maintenance.
- Solid knowledge of SDLC, worked as team leader using Agile Scrum software development methodologies.
- Mobile platform software development using Android, Google Kotlin, and Ionic multi-platform “Android, iOS, and PWA” development framework.
- Experience in programming using JavaScript, Node.js and Angular, React Frameworks, with working knowledge of Python using Django, Flask and FastAPI.
- Experience OOP using Java programming language using Spring framework.
- Over 10 years of experience in using relational database such as PostgreSQL, MySQL, and Oracle DB.
- Solid knowledge and some experience in NoSQL and multidimensional databases such as MongoDB and Casandra DB.
- Experienced in using Git and Gitlab and containerized deployments using Docker and Kubernetes orchestration.
- Experience dealing with stakeholders and product owners. Highly organized and punctual, with strong work ethics. Keen to learn new skills and technologies.
- Highly adaptable to new environments, can work independently or as part of a collaborating team, and able to communicate fluently in both English and Arabic.

EDUCATION

PhD. in Computer Science University of New Brunswick, Fredericton, NB, Canada Thesis title is “ <i>Design Pattern as a Service for Service-Oriented Systems</i> ”	Sep. 2019
MSc. in Computer Science University of New Brunswick Fredericton, NB, Canada	Oct. 2003
Bachelor of Computer Science Sebha University, Sebha, Libya	Apr. 1990

TECHNICAL EXPERTISE

Software Technologies: Android mobile app development, Desktop and Client/Server Multi-tier Architectures, Software Patterns, Object-Oriented Programming (OOP), RESTful APIs, Software as a Service (SaaS), Cloud Computing, Microsoft Azure.

Programming Languages and Frameworks: Ionic framework, Google Kotlin, Angular, React, Java, J2EE, Java Servlets, Java Server Pages (JSP), java Portlets, JavaScript, JSON, Python, Django, Flask, FastAPI, ASP.NET, C#, C++, low-code Noe-Red, MQTT, HTML5, Bootstrap, CSS, XML, XML Schema Definition (XSD).

Databases: SQL Server, MySQL, PostgreSQL, Oracle DB, Mongo and Cassandra no-SQL multi-dimensional database (MDD).

PROJECTS

NBCC Applied Research Projects

Jul. 2020 – Dec.2025

- Research study for Hotspot parking inc. <https://www.htsp.ca>. In this study I researched the perspectives for the company to enter the U.S. parking market and compared and contrasted the use of SMS Text-to-pay paradigm with the QR code based system. The study covered the statistics for user adoption and convergence of the two systems and the security risks and good practices. The study closes with some recommendations of some SaaS platforms with the easiest integration procedures and the most well-documented APIs. Nov. 2025 – Dec. 2025
- Lead the Information and Communications Technology Research Group (ICTRG) of the College Office of Research Enterprise (CORE) at New Brunswick Community College (NBCC) to design and develop an Interactive Data Layer (IDL). This project created a common data layer for all IoT sensor belonging to several research projects. The IDL collected data from those IoT sensors and exposed it for analysis tools and decision support software as well as providing real-time dashboards for time-series data. In addition to that, the IDL offered a group of API end points for third-party applications to access and utilize the data housed by the IDL. Nov. 2024 – Oct. 2025
- Developed and deployed an IoT sensors project to collect environmental data from the E-Building in NBCC's Saint John Campus. This project is in collaboration with Saint John Energy company. Several Environmental sensors for measuring air quality in the building correlated with the number of people inside the building and offer a dashboard to show real-time Air Quality Index (AQI) status of the building and issue recommendations. Apr. 2024 – Oct. 2024
- Development of an IoT system that collects data from a group of IoT sensors deployed in Fredericton Campus of the New Brunswick Community College (NBCC). The collected data includes temperature humidity, light intensity, noise levels as well as counting people inside the building. All this data is fed to a recommendation model for neurodivergent users of the campus. The technologies used are IoT sensors, LoRaWAN, Chirpstack software platform for managing IoT sensor data sent by the LoRaWAN gateway. I also built an API with multiple endpoints for Black Arcs to consume the IoT data and use it to build a digital twin of the college campus. Jun. 2023 – Mar. 2024
- Team member of an agricultural IoT research project. The project designs and implements a system for the early detection of apple scab disease. The system uses weather stations and leaf wetness sensors and utilizes LoRaWAN (Long Range Radio Wide Area Network) communication technology to wirelessly connect the sensors to an edge server. I implemented the system and the dashboard for apple scab monitoring using JavaScript, Node.js and PostgreSQL database. Jun. 2022 – May 2023
- Research and development of a mobile application for both iOS and Android platforms using Ionic framework to manage and interact with Low-Energy Bluetooth (BLE) devices built by Greystone Energy Systems (GES), Moncton, NB. Jul. 2020 – Jun.2022

NSERC Research Project

Sep. 2013 – Jun. 2014

- Integrated CrowdBucks Trading Network (CTN) with several software platforms and services such as Cyclos as a banking backend, Liferay portal as a frontend, Paypal as a payment system, social networks, e-commerce and e-marketplace applications. All these applications and platforms were integrated using Mulesoft Enterprise Service Bus (ESB) as a middleware layer.

Automated a remote sensing project

Feb. 2013 – Apr. 2013

- Using ArcGIS, R, and Python - the automation project reads data from satellite sensors, transforms the data into more suitable formats for processing, extracts interesting pieces of data, and uses that data to identify buildings in the area being surveyed from other dense areas such as vegetation, bushes, and trees.

NSERC Project to enhance performance of SHIFT Energy Software system Jan. 2012 – Jun. 2012

- Investigated the benefits of using multi-dimensional databases to enhance the performance of SHIFT Energy's software system and built a prototypical system as part of the study using Java and NoSQL Cassandra multi-dimensional database.

WORK EXPERIENCE

Researcher | Applied Research, NBCC, Fredericton, NB, Canada. Jul. 2020 – Dec. 2025

Instructor (PT) | Faculty of Computer Science, UNB, NB, Fredericton, Canada. Apr. 2016 – Jun. 2020

- **Course:** CS1083 Introduction to Computer Programming II (in Java): Develop lesson plans, conduct teaching sessions, preparing lab material, assignments, and assessment procedures for computer science undergraduate students. Guiding and offering academic advice and tutoring.

Assistant Lecturer | Faculty of Science, Aljabal Algharbi University, Zintan, Libya Apr. 2006 – Jun. 2012

- **Courses:** Introduction to Computer Science, Operating Systems, File Processing, Principles of Software Engineering, Object Oriented Architecture and Programming using Java, and C/C++ Programming Languages:
- Developed an instructional plan as a course outline or Syllabus. Planned lessons, assignments, and quizzes. Worked with colleagues to develop or modify the curriculum for a degree or certificate program involving a series of courses.
- Assessed students' progress by grading papers, tests, examinations, and other work.
- Advised students with their course load and goals. Supervised graduation projects for undergraduate students.

Systems Analyst and Programmer, Senior Systems Analyst and Programmer, Systems Specialist, and Senior Systems Specialist | Waha Oil Company, Tripoli, Libya. Aug. 1990 – Mar. 2006

- Analyzed, designed, developed, upgraded, and maintained of the company's Inventory, Accounting, Payroll, and Oil Movement systems. Member of the team that upgraded the company's Inventory system to avoid the Y2K problem. Member of the company's Intra-Net project team which developed an in-house ERP system for the company.
- Produced weekly, biweekly, and monthly reports for Accounting, Payroll, Materials, and Operations departments. Performed periodic data integrity routines.
- Performed disaster recovery procedures to recover from system crashes. Supervised and mentored junior Systems analysts and programmers.

PROFESSIONAL COURSES AND WORKSHOPS

Professional Grant Development Workshop granttrainingcenter.com	Dec. 2022
Full Stack Web Development (HTML, CSS, React, and Node) Udemy.com	May 2020
Time Management Mitacs Step Fredericton, NB, Canada	Apr. 2015
Foundations of Management I & II Mitacs Step Fredericton, NB, Canada	Mar. 2015
Practice Your Presentation Skills I Mitacs Step Fredericton, NB, Canada	Mar. 2015
Skills of Communication Mitacs Step Fredericton, NB, Canada	Feb. 2015